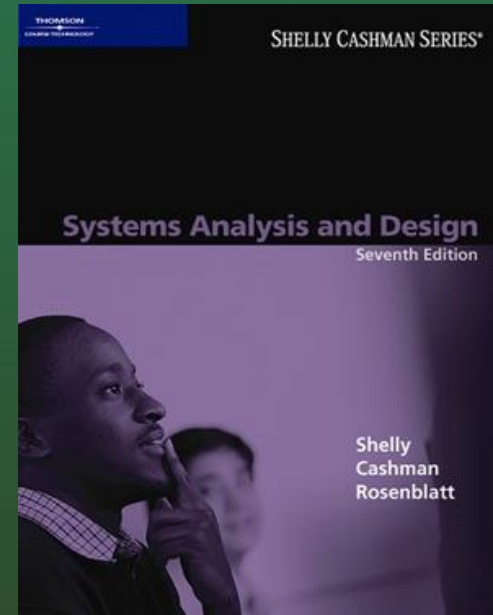


Systems Analysis & Design 7th Edition

Chapter 2



Phase Description

- Systems planning is the first of five phases in the systems development life cycle (SDLC)
- In this phase, you will learn how IT projects get started and how a systems analyst evaluates a proposed project and determines its feasibility



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

MODULE 2

LEARNING OBJECTIVES

Chapter Objectives

- Explain the concept of a business case and how a business case affects an IT project
- Describe the strategic planning process and why it is important to the IT team
- Explain the purpose of a mission statement
- Describe the SDLC, and explain how it serves as a framework for systems development and business modeling
- Describe risks and risk management features
- List the reasons for information systems projects and the factors that affect such projects



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Chapter Objectives

- Explain the initial review of systems requests and the role of the systems review committee
- Define operational feasibility, technical feasibility, economic feasibility, and schedule feasibility
- Describe the steps in a preliminary investigation and the end product of an investigation



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Introduction

- The term business case refers to the reasons, or justification, for a proposal
- Systems development typically starts with a systems request, followed by a preliminary investigation, which includes a feasibility study



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Strategic Planning – A Framework for IT Systems Development

- Strategic planning is the process of identifying long-term organizational goals, strategies, and resources
- Strategic Planning Overview
 - SWOT analysis



FEU ALABANG



FEU DILIMAN

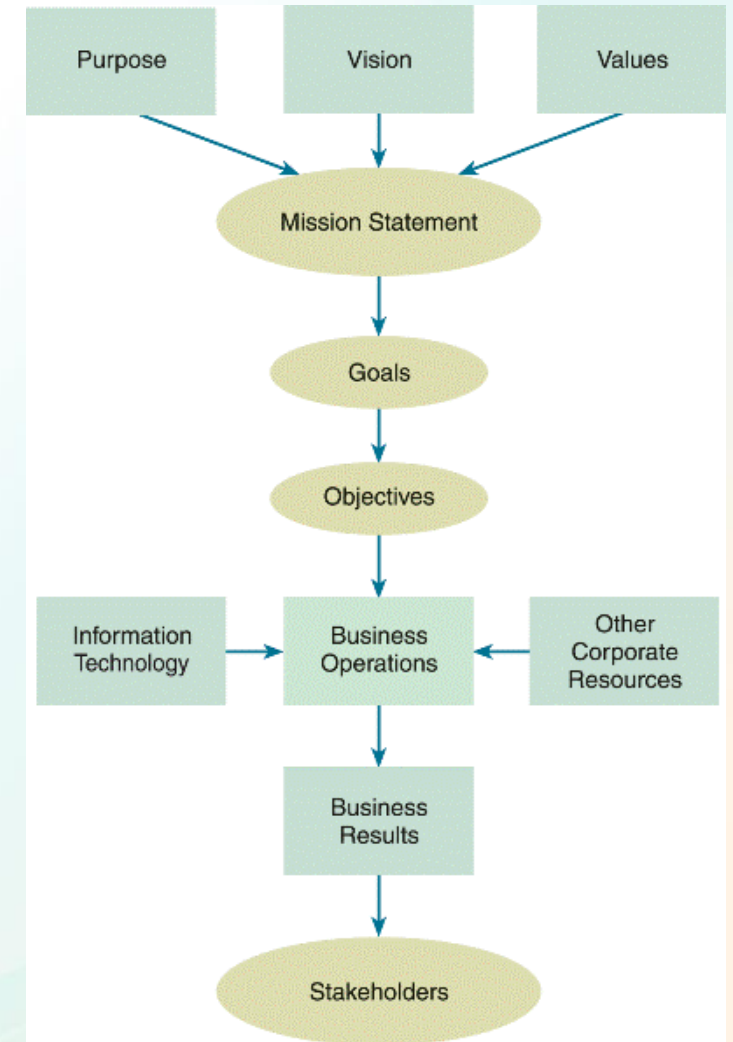


FEU TECH

Technology Driven by Innovation

Strategic Planning – A Framework for IT Systems Development

- From Strategic Plans to Business Results
 - Mission statement
 - Stakeholders
 - Goals
 - Objectives



Strategic Planning – A Framework for IT Systems Development

- A Business Example
 - Critical success factors
 - Critical business issues
 - Case for action



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Strategic Planning – A Framework for IT Systems Development

- The Role of the IT Department in Project Evaluation
 - Management leadership and information technology are linked closely, and remarkable changes have occurred in both areas
 - Today, systems development is much more team oriented
 - Although team-oriented development is the norm, some companies see the role of the IT department as a gatekeeper



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Strategic Planning – A Framework for IT Systems Development

- The Future
 - If you could look into the future, here is what you might see: New industries, products, and services emerging from amazing advances in information technology, customers who expect world-class IT support, a surge in Internet-based commerce, and a global business environment that is dynamic and incredibly challenging



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

What Is a Business Case?

- Should be comprehensive, yet easy to understand
- Should describe the project clearly, provide the justification to proceed, and estimate the project's financial impact

Information Systems Projects

- Main Reasons for Systems Projects
 - Systems request
 - Improved service
 - Support for new products and services
 - Better performance
 - More information



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Information Systems Projects

- Main Reasons for Systems Projects
 - Stronger controls
 - Encryption and biometric devices
 - Reduced cost
- Factors that Affect Systems Projects
 - Internal and external factors affect every business decision that a company makes, and IT systems projects are no exception



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Information Systems Projects

- Internal Factors
 - Strategic plan
 - Top managers
 - User requests
 - Information technology department
 - Existing systems and data



FEU ALABANG



FEU DILIMAN

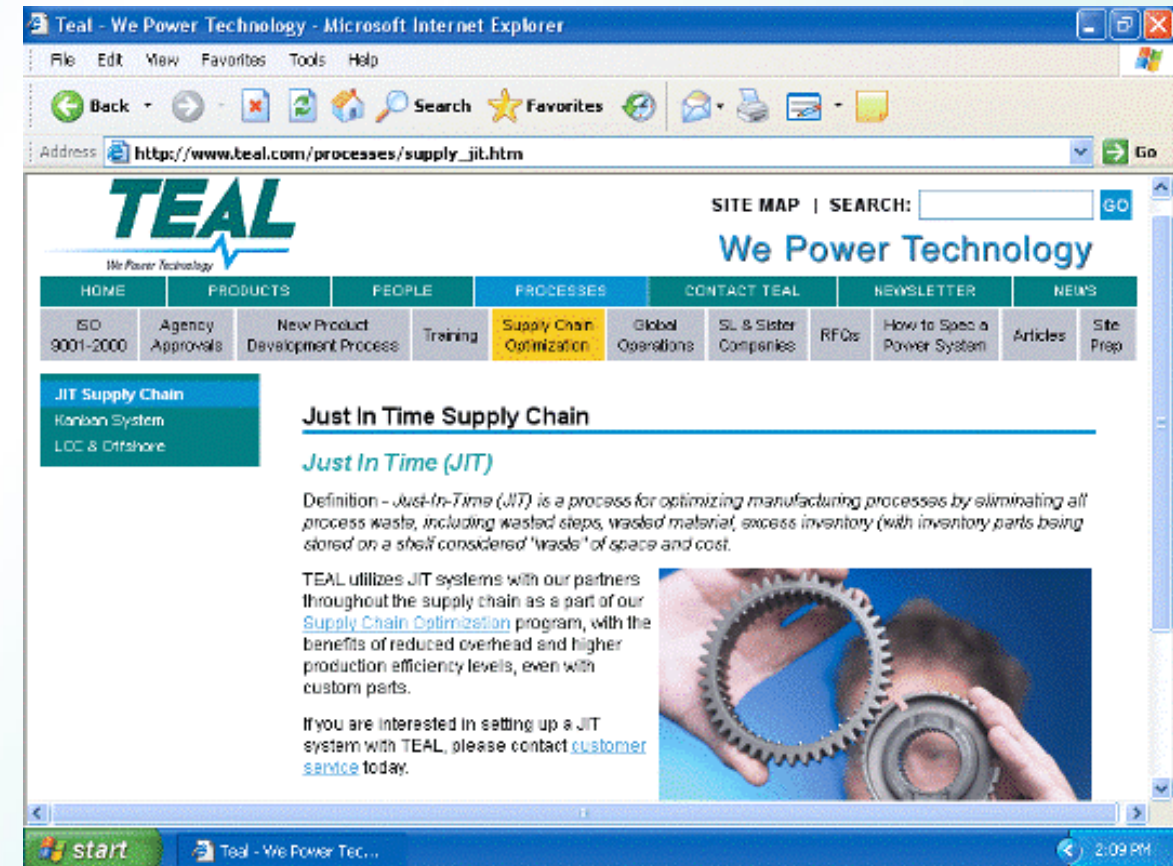


FEU TECH

Technology Driven by Innovation

Information Systems Projects

- External Factors
 - Technology
 - Electronic product code (EPC)
 - Suppliers
 - Just-in-time (JIT)



Information Systems Projects

- External Factors
 - Customers
 - Customer Relationship Management (CRM)
 - Electronic proof of delivery (EPOD)
 - Competitors
 - The economy
 - Government



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Information Systems Projects

- Project Management Tools
 - All IT projects, large and small, must be managed and controlled
 - Project management begins with a systems request, and continues until the project is completed or terminated
- Risk Management
 - Every IT project involves risks that system analysts and IT project managers must address
 - Risk management



FEU ALABANG



FEU DILIMAN

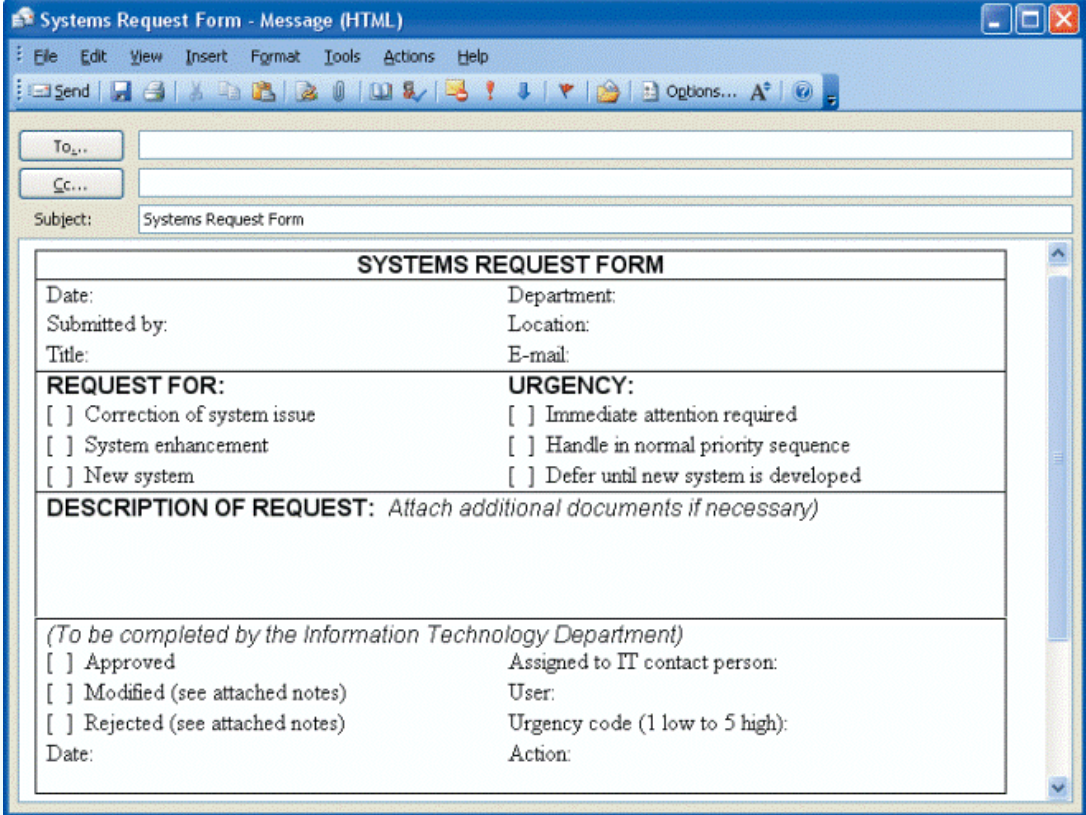


FEU TECH

Technology Driven by Innovation

Evaluation of Systems Requests

- Systems review committee
- Computer resources committee
- Systems Requests Forms



The screenshot shows a web browser window titled "Systems Request Form - Message (HTML)". The browser's address bar is empty, and the subject line of the message is "Systems Request Form". The form itself is titled "SYSTEMS REQUEST FORM" and contains the following fields and sections:

Date:	Department:
Submitted by:	Location:
Title:	E-mail:
REQUEST FOR:	
☐ Correction of system issue	☐ Immediate attention required
☐ System enhancement	☐ Handle in normal priority sequence
☐ New system	☐ Defer until new system is developed
URGENCY:	
DESCRIPTION OF REQUEST: <i>Attach additional documents if necessary</i>	
<i>(To be completed by the Information Technology Department)</i>	
☐ Approved	Assigned to IT contact person:
☐ Modified (see attached notes)	User:
☐ Rejected (see attached notes)	Urgency code (1 low to 5 high):
Date:	Action:



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Evaluation of Systems Requests

- Systems Review Committees
 - Most large companies use a systems review committee to evaluate systems requests
 - Many smaller companies rely on one person to evaluate system requests instead of a committee
 - The goal is to evaluate the requests and set priorities



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Overview of Feasibility

- A systems request must pass several tests, called a feasibility study, to see whether it is worthwhile to proceed further



Overview of Feasibility

- Technical Feasibility
 - Technical feasibility refers to technical resources needed to develop, purchase, install, or operate the system
- Economic Feasibility
 - Total cost of ownership (TCO)
 - Tangible benefits
 - Intangible benefits
- Schedule Feasibility



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Evaluating Feasibility

- The first step in evaluating feasibility is to identify and weed out systems requests that are not feasible
- Even if the request is feasible, it might not be necessary
- Feasibility analysis is an ongoing task that must be performed throughout the systems development process

Setting Priorities

- Factors that Affect Priority
 - Will the proposed system reduce costs? Where? When? How? How much?
 - Will the system increase revenue for the company? Where? When? How? How much?



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Setting Priorities

- Factors that Affect Priority
 - Will the systems project result in more information or produce better results? How? Are the results measurable?
 - Will the system serve customers better?
 - Will the system serve the organization better?

Setting Priorities

- Factors that Affect Priority
 - Can the project be implemented in a reasonable time period? How long will the results last?
 - Are the necessary financial, human, and technical resources available?
 - Whenever possible, the analyst should evaluate a proposed project based on tangible costs and benefits that represent actual (or approximate) dollar values



FEU ALABANG



FEU DILIMAN



FEU TECH

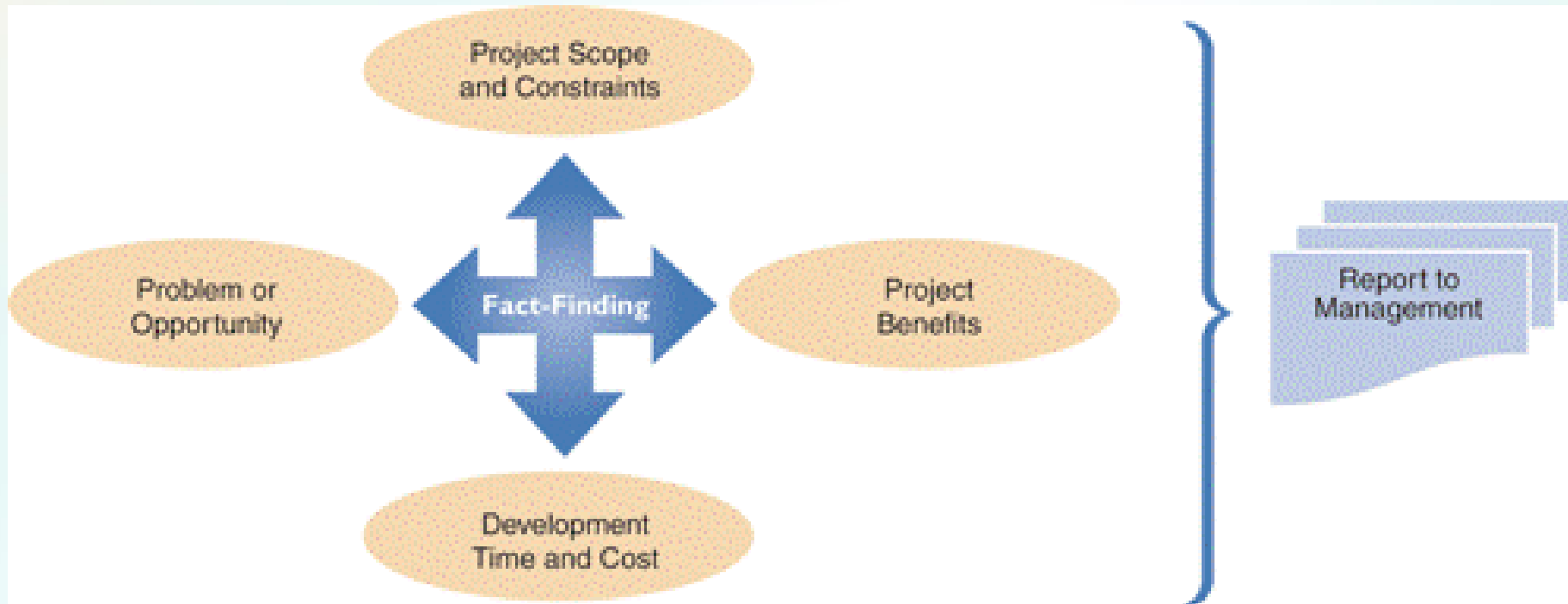
Technology Driven by Innovation

Setting Priorities

- Discretionary and Nondiscretionary Projects
 - Projects where management has a choice in implementing them are called discretionary projects
 - Projects where no choice exists are called nondiscretionary projects

Preliminary Investigation Overview

- Preliminary investigation
- Interaction with Managers and Users



Preliminary Investigation Overview

- Planning the Preliminary Investigation
 - During a preliminary investigation, a systems analyst typically follows a series of steps
 - The exact procedure depends on the nature of the request, the size of the project, and the degree of urgency



FEU ALABANG



FEU DILIMAN

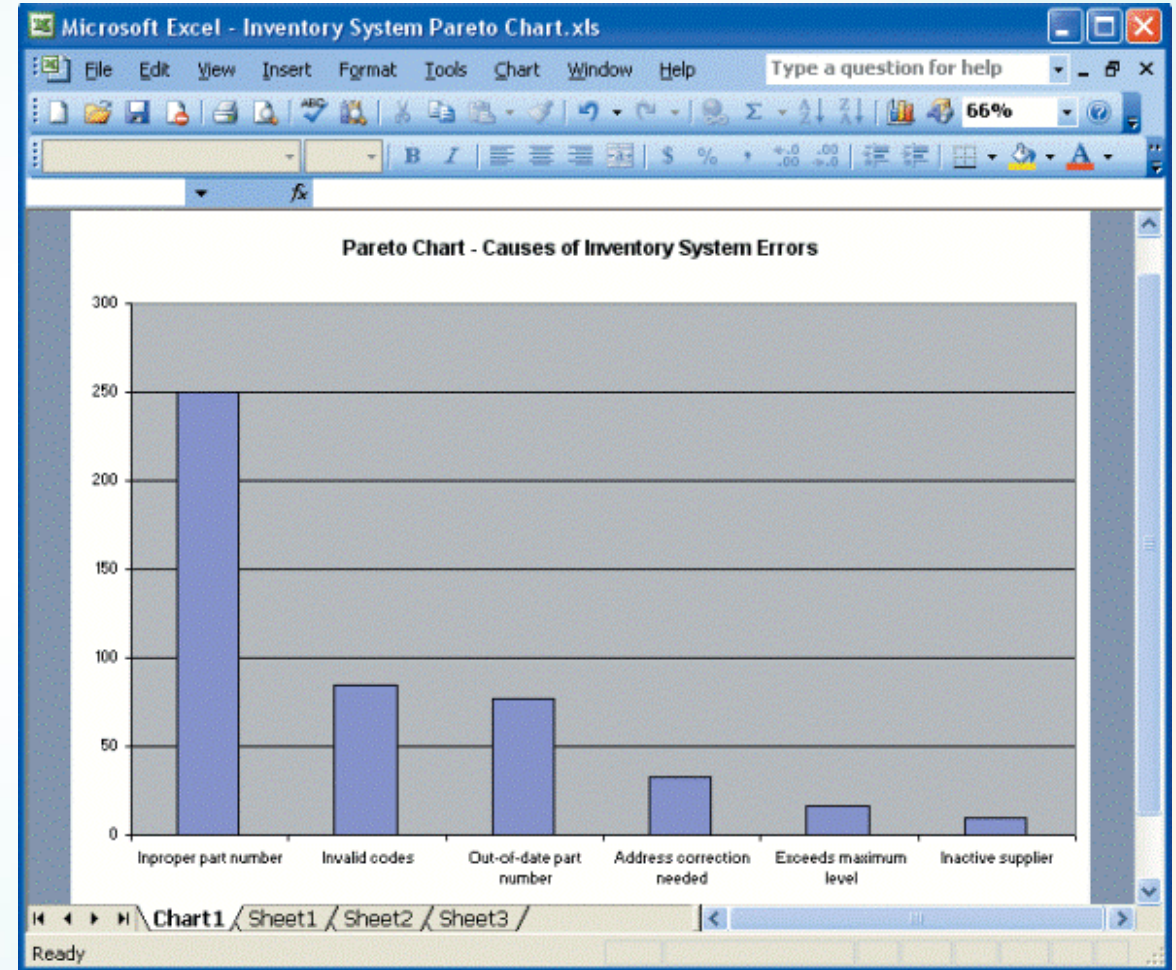


FEU TECH

Technology Driven by Innovation

Preliminary Investigation Overview

- Step 1: Understand the Problem or Opportunity
 - A popular technique for investigating causes and effects is called a fishbone diagram, or Ishikawa diagram
 - Pareto chart



Preliminary Investigation Overview

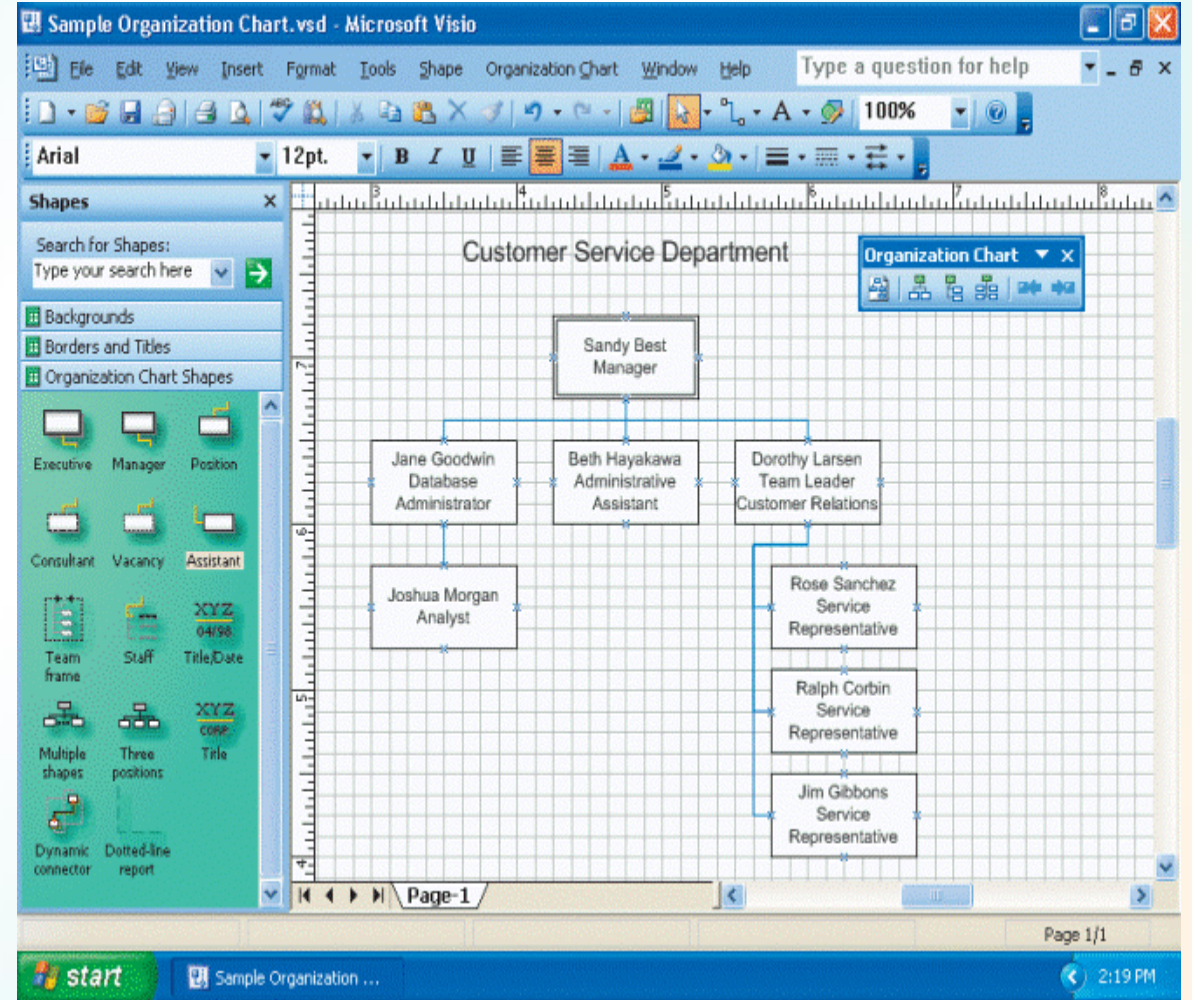
- Step 2: Define the Project Scope and Constraints
 - Project scope
 - Project creep
 - Constraint
 - Present versus future

Preliminary Investigation Overview

- Step 2: Define the Project Scope and Constraints
 - Present versus future
 - Internal versus external
 - Mandatory versus desirable
 - Regardless of the type, all constraints should be identified as early as possible to avoid future problems and surprises

Preliminary Investigation Overview

- Step 3: Perform Fact-Finding
 - Fact-finding involves various techniques
 - Fact-finding might consume several hours, days, or weeks
 - Analyze Organization Charts



Preliminary Investigation Overview

- Step 3: Perform Fact-Finding
 - Conduct interviews
 - Determine the people to interview
 - Establish objectives for the interview
 - Develop interview questions
 - Prepare for the interview
 - Conduct the interview
 - Document the interview
 - Evaluate the interview

Preliminary Investigation Overview

- Step 3: Perform Fact-Finding
 - Review documentation
 - Observe operations
 - Conduct a user survey
- Step 4: Evaluate Feasibility
 - Evaluate the project's operational, technical, economic, and schedule feasibility



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Preliminary Investigation Overview

- Step 5: Estimate Project Development Time and Cost
 - What information must you obtain, and how will you gather and analyze the information?
 - What sources of information will you use, and what difficulties will you encounter in obtaining information?

Preliminary Investigation Overview

- Step 5: Estimate Project Development Time and Cost
 - Will you conduct interviews? How many people will you interview, and how much time will you need to meet with the people and summarize their responses?
 - Will you conduct a survey? Who will be involved? How much time will it take people to complete it? How much time will it take to prepare it and tabulate the results?



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Preliminary Investigation Overview

- Step 5: Estimate Project Development Time and Cost
 - How much will it cost to analyze the information gathered and to prepare a report with findings and recommendations?
 - You should provide an estimate for the overall project, so managers can understand the full cost impact and timetable
- Step 6: Present Results and Recommendations to Management
 - The final task in the preliminary investigation is to prepare a report to management
 - The format of the preliminary investigation report varies from one company to another



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

Preliminary Investigation Overview

- Step 6: Present Results and Recommendations to Management
 - Introduction
 - Systems request summary
 - Findings
 - Recommendations
 - Project Roles
 - Time & cost estimates
 - Expected benefits
 - Appendix



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

CHAPTER SUMMARY

Chapter Summary

- Systems planning is the first phase of the systems development life cycle
- Effective information systems help an organization support its business process, carry out its mission, and serve its stakeholders
- Strategic planning allows a company to examine its purpose, vision, and values and develops a mission statement, which leads to goals, objectives, day-to-day operations, and business results that affect company stakeholders
- Systems projects are initiated to improve performance, provide more information, reduce costs, strengthen controls, or provide better service



Chapter Summary

- Various internal and external factors affect systems projects
- During the preliminary investigation, the analyst evaluates the systems request and determines whether the project is from an operation, technical, economic, and schedule standpoint
- Analysts evaluate systems requests on the basis of their expected costs and benefits, both tangible and intangible
- The steps in the preliminary investigation are to understand the problem or opportunity; define the project scope and constraints; perform fact-finding; estimate the project's benefits; estimate project development time and cost; and present results and recommendations to management



FEU ALABANG



FEU DILIMAN



FEU TECH

Technology Driven by Innovation

